

TEZPUR UNIVERSITY

Spring Semester MAJOR I, 2022

Design and Analysis of Algorithms (CO 206)

Maximum Marks : 40

Duration : 2 hours

You may answer all the questions but maximum you can get is 40.

1. For each of the following statements indicate whether the statement is true or false and briefly explain why. [5 × 4]

a) The time complexity of the following code is $O(n)$.

```

1  for (i=n; i>0; i=i/2){
2      for (j=0; j<i; j++){
3          ..... // O(1) operations
4      }

```

b) $2^{2n} \in O(2^n)$.

c) In the recursive closest pair algorithm for N points, total number of calls to **AcrossPair()** is $O(N)$. (Recall that **AcrossPair()** is called on subarrays of length 2 or more to find the closest pair that are on different sides of the dividing line.)

d) The number of pivots selected throughout the Quicksort algorithm to sort an array of n elements is $\Omega(\log n)$.

e) Among any 4 integer numbers, there will be 2 numbers whose difference is divisible by 3.

2. Write a recurrence representing the runtime of the method below. Compute the running time, and analyze the worst case running time using $O(\cdot)$ notation. [10]

```

1  AlgoAK(int n)
2  {
3      if (n % 2 != 0) {
4          for (i=1; i<=n, i++)
5              result = result + i;
6          return 2 * AlgoAK(n/5);
7      }
8      else
9      {
10         for (i=1; i<=n, i++)
11             result = result + i;
12         return AlgoAK(n/2);
13     }
14 }

```

3. Recall that merge sort works by taking an array of n elements, splitting it into two subarrays of equal size, i.e., $\frac{n}{2}$, recursively sorting both subarrays, then combining both of them. Suppose we split the array into two subarrays of unequal sizes. The size of the left and right ones are $\frac{2n}{7}$ and $\frac{5n}{7}$ respectively. And after we finish recursively sorting each of the two subarrays, we merge them together to get the final sorted result.

- a) Compute the recurrence for the runtime of the above variation of MergeSort. What will be the base case? [5+2]
- b) Compare the asymptotic runtime of this variation with that of the standard merge sort. Is this variation better, the same, or worse than the standard merge sort? Justify your answer logically. [5+5]
4. At a ticket counter, n persons are waiting in a queue each of whom is either MALE or FEMALE or TRANSGENDER. Moreover, assume that there is at least one person of each gender and that all persons of the same gender are standing contiguously. Each person has a token mentioning the serial number of that person. E.g., the 1st person in the queue has a token number 1, last one has token number n etc. Design an efficient algorithm to determine the number of persons of each gender. You may perform only two operations on a person; You can ask question of the following two kinds:
- a) What is your gender?
- b) What is your token number? [5+5]

The only information you have is the value of n . You have to count the number of persons in each category by asking questions to the individuals. Your algorithm requires to find the solution by asking fewest number of questions as possible. Determine in $O()$ notation how many questions are sufficient to solve the above problem? [10]